

IT team mail – BIC

Subject: Awareness & Technical Guidelines for the Protection of Sensitive Information

To: BIC IT Team

From: Information Security Team

Hello IT Team,

We know that you are already engaged daily in protecting our systems and securing information flows. Today, we would like to strengthen together an essential approach: the classification, storage, sharing, and archiving of BIC's sensitive data, while integrating best practices in cryptography and cybersecurity.

1. Classification and Responsibilities

The first step is to clearly distinguish public data (freely communicated externally) from confidential data (internal, strategic, or sensitive). Confidential data, in particular, must always be protected through appropriate mechanisms: encryption, strict access management, and secure archiving.

2. Cryptographic Best Practices

- **Asymmetric encryption (RSA / ECC):** to be prioritized for all external exchanges, ensuring a clear separation between public keys (shareable) and private keys (strictly protected).
- **Symmetric encryption (AES-256):** to be used for internal storage and archiving of large volumes of data, guaranteeing both performance and security.
- **Secure hashing (SHA-256 / SHA-3):** to ensure the integrity of sensitive files, allowing detection of any unauthorized modification.
- **Key management:** storage in HSMs (Hardware Security Modules) or digital vaults, regular key rotation, and immediate revocation in case of suspected compromise.

3. Password Security

Passwords often represent the first line of defense.

- **Recommendations:** minimum of 12 characters for common uses and 16 characters minimum for sensitive access.
- Favor passphrases (long sentences, easy to remember but hard to guess).
- Avoid classic patterns (capital letter at the beginning, digits at the end).
- Enable multi-factor authentication (2FA) whenever possible to reduce the impact of a compromised password.

4. Common Attacks

- **Brute force:** testing all possible combinations.
- **Dictionary:** using known passwords or predictable patterns.
- **Credential stuffing:** exploiting credentials already compromised in other breaches.

These attacks are automated, fast, and massive. To protect against them, it is essential to combine strong passwords, secure hashing mechanisms, MFA, and strict access controls.

5. Password Hashing and Storage

A password must never be stored in cleartext. Best practices include:

- **Secure hashing functions:** SHA-256, SHA-3, or even better, algorithms specifically designed for password hashing (bcrypt, Argon2).
- **Salting:** each password must be accompanied by a unique salt per user, to neutralize precomputed table (rainbow table) attacks.
- **Iterations (key stretching):** increasing computational workload to slow down brute force attacks.

This way, even if a database is compromised, passwords remain difficult to exploit.

6. Secure Storage and Sharing

- Internal repositories must be configured with encryption at rest and in transit (TLS 1.3 required).
- Removable media (USB drives, external disks) must not contain unencrypted sensitive data.
- The use of external cloud solutions can only be tolerated if they integrate end-to-end encryption with company-controlled key management.

7. Archiving and Traceability

- Any archived confidential data must be protected through strong encryption, accompanied by logging to trace access.
- Retention duration must align with internal data retention policy and the information lifecycle.

8. Your Key Role

You are the technical guardians of this trust chain. Your expertise is essential to ensure the confidentiality, integrity, and availability of BIC's data at all times.

We are counting on you to:

- Systematically apply these practices,
- Train and raise awareness among non-technical teams when they seek your assistance,
- Report any incident or suspicion of compromise.

Conclusion : Information security is a collective challenge that relies both on robust technical solutions and responsible behavior. By applying these principles of cryptography, password management, encryption, and secure archiving, we strengthen the long-term protection of our sensitive data and ensure the continuity of the company's operations.

We thank you for your commitment. Your rigor and sense of responsibility are essential for the company to continue moving forward safely.

Please do not hesitate to reach out to us if you have any questions.

Kind regards,

The Information Security Team,

BIC

